

CAN7032/7132 Computational genomics, transcriptomics and evolution, 2025/2026
Coursework assignment

Submission deadline: 4pm of Thursday 26th Feb 2026 (UK time)

This coursework assignment consists of 2 short answer questions (SAQs) and 4 long answer questions (LAQs). SAQs are to test your knowledge and understanding of the topics covered by this module, and LAQs are to assess your analytic skill and abilities.

SAQ 1: Briefly describe the characteristics of the cancer genome (10 marks)

SAQ 2:

- 1. What is cancer immunotherapy? (2 marks)**
- 2. Give two examples of cancer immunotherapy, and also brief describe the underlying mechanism of each immunotherapy? (4 marks)**
- 3. Briefly describe strategies that tumour cells take to evade immune surveillance. (4 marks)**

LAQ 1:

High grade serous ovarian cancer cell lines Kuramochi and Snu119 were single cell sorted, such that one individual cell was dispensed into one well of a 96 well plate to form a new clonal population (a 'clone'). Each new clone was expanded into a new clonal population over 22 cell division cycles. Single cell DNA sequencing (low pass whole genome sequencing) was performed on small subsets of (i) the parental population that the single cell was derived from, and (ii) the new clonal populations (C8 & E10 (Kuramochi) and A12 & F12 (Snu119)). **(Total 25 marks)**

Get the data required for this question from `"/data/teaching/bci_teaching/Computational_genomics/".` You can find further instructions on how to transfer the data to your hpc folder in the slide no 8 in ['Practical 2_Study Tumour Evolution using single cell sequencing_2026.pptx'](#) file.

- 1. 5 marks** are available for successfully running the data through Aneufinder and MEDICC2 and selecting the relevant data to contribute to the following questions.
- 2. Which type of alterations can you identify in the single cell whole genome sequencing data that is output from AneuFinder from the ovarian cancer genomes? Which types of alteration(s) cannot be identified in these profiles, and why? (4 marks)**

3. Calculate the number of genomic breakpoints (location where copy number change is detected) per cell and per chromosome from the AneuFinder browser CNV files. Are the CNAs distributed evenly across the chromosomes? Demonstrate this using representative figures or graphs. **(5 marks)**
4. What is the size range of CNAs across the two HGSOC cell lines and clones and how often are these whole chromosome alterations vs sub-chromosomal? Demonstrate this using representative figures or graphs. **(4 marks)**.
5. Using metrics produced by MEDICC2, calculate the total branch lengths per cell from the branch_lengths.tsv (be careful to remove first internal branch which indicates the length from diploid to the ancestral cancer cell). Produce scatter plots for the clone populations of Kuramochi & Snu119. Comment on the significance of the branch lengths of the populations and how this could be informative for inferring rates of genomic alteration, given that all clones were grown for 22 cell divisions **(7 marks)**.

LAQ 2:

In the CAN7031 “omics analytics” coursework assignment, you were given the raw count matrix of 10 matched normal and tumour samples from 5 patients, S02, S03, S05, S06 and S10. Using the normalised gene expression matrix you created in the task (e.g., RPKM or CPM log2 transformed or not transformed) to perform the bulk RNA deconvolution analysis. You are required to answer the following questions. **(Total 20 marks)**

- (1) Use two individual and one consensus methods to perform the bulk RNA deconvolution. And produce heatmaps based on the cellular estimates, with samples clustered based on the profile. Briefly describe the patterns you can see. (Hint: pay attention to the input format depending on the methods you use, e.g., logged or unlogged values) **(8 marks)**
- (2) Perform student t-test on the cellular estimates between tumour and normal groups for these estimates from the three methods above. List the cell types with the significant differences ($p < 0.05$), and produce boxplots for these significant ones. **(5 marks)**
- (3) Perform the GSVA analysis (i.e., tumour vs. normal) using the KEGG genesets provided in the practical session, and generate a heatmap for mostly significantly differentially regulated KEGG pathways, using the cutoff of adjusted p-value ≤ 0.01 . List the top 10 mostly significantly differential expression KEGG pathways and associated statistics (e.g., fold change, p-value, adjusted p-value etc) in a table. **(7 marks)**

LAQ 3:

In the data folder under the “Assessments” tab, you will find the VCF files of four skin tumour samples, namely WD06, WD13, WD14, WD22, with somatic mutations called against the matched normal reported with the coordinates **in the hg38 format**.

Perform the analysis of tumour neoantigen analysis using MuPeXi, selecting two major HLA-A alleles, HLA-A02:01, HLA-A24:02, and peptide length of 8-11 mers. No need to provide the expression file. **(Total 18 marks)**

- (1) How many tumour neoantigens can you find for each tumour? How many strong tumour neoantigens are there for each tumour? (Hint: use the rank score cutoff values mentioned in the lecture/practical). **(7 marks)**
- (2) For all the strong tumour neoantigens identified, what are the affected genes? What are the neoantigen peptides? What are the rank scores in tumour and normal, respective, for these strong tumour neoantigens? Create a table for all the strong tumour neoantigens, with all the essential statistics, e.g., including rank scores, and with one column indicating the tumour sample. **(7 marks)**
- (3) Are there any shared tumour neoantigens amongst the four tumour samples? What are the potential implications if neoantigen-based therapies are designed? **(4 marks)**

LAQ 4:

MClust tumour clonality analysis **(Total 17 marks)**.

In the data folder under the “Assessments” tab, you will find “**LU-4.snv.txt**” that contains all somatic mutations identified for the LU-4 tumour sample, and “**mutations_SP4_LN1_LN2_together.txt**” that contains all mutations identified and shared between SP4 lymph node 1 (LN1) and lymph node 2 (LN2) tumour samples. This question is to investigate the clonal evolution patterns for a single sample “LU-4”, and for comparing two sites, LN1 vs. LN2, of the SP4 patient, using the mClust analysis.

For the single sample analysis,

- (1) Load the “LU-4.snv.txt” file, and run the mClust single sample clustering analysis using variant allele frequency (VAF) of all mutations identified. First, generate a histogram of all VAFs, and briefly comment on the VAF distribution **(2 marks)**.
- (2) Run the density estimation analysis via Gaussian finite mixture modelling, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF)? You can summarise all these into a table as shown in the lecture/practical slides **(3 marks)**.
- (3) Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specify clearly the cluster names/numbers and associated CCF values **(2 marks)**.

For the paired sample analysis,

- (4) Load the “mutations_SP4_LN1_LN2_together.txt” file, and run the mClust paired sample analysis using VAFs of all shared mutations between SP4 LN1 and LN2 samples. First, generate a scatter plot of all VAFs between LN1 and LN2 samples, and briefly comment on the plot on shared mutations between them (**2 marks**).
- (5) Run mClust analysis based on Gaussian finite mixture model fitted by EM algorithm, followed by dimension reduction clustering and classification analysis. How many clusters can you identify? What is the size (# of mutations) for each cluster? What is the mean VAF value for each cluster in LN1 and LN2 samples, respectively? Can you identify which are major clones (heterozygous or homozygous), which are subclones, and their associated cancer cell fractions (CCF) in LN1 and LN2 samples, respectively? You can summarise all these into a table as shown in the lecture/practical slides (**4 marks**).
- (6) Finally, please draw a tumour evolutionary tree to illustrate the clustering patterns listed above. Within the tree, specify clearly the cluster names/numbers and associated CCF values in LN1 and LN2 samples, respectively when applicable (**4 marks**).